Towards Characterizing Solar System Networking: Analyzing the DSN Now Data Set

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Abstract

Extending the Internet into space is becoming an increasingly prominent area of research and development. Limited empirical evidence of the unique characteristics of deep space communication, however, challenges the successful design of architectures and protocols for non-terrestrial networks to implement an interplanetary Internet. In this short paper, we provide the first analysis of live telemetry data from the NASA Deep Space Network (DSN). We systematically characterize latency, data rate, and connectivity over a period of eleven months. We found several unexpected results. First, nearly half of communication links operate with subminute one-way latencies, challenging the common belief that space communication always involves extreme delays. Second, one-fifth can be attributed to spacecraft positioned at Sun-Earth Lagrange points, which are often overlooked despite their critical role in space communication. Third, data rates vary substantially across deep space missions, exhibiting a weaker correlation with distance than anticipated. Even with a globally distributed ground station network, Earth rotation and scheduling constraints cause extended signal disruptions, while dish tracking issues result in frequent disruptions lasting several minutes. We release the archived data set to establish an empirical foundation for deep space network simulation, protocol research, and engineering.

1 Introduction

Networking in space poses distinct challenges that are not present in terrestrial networks. High latencies are common, with the round-trip time for a signal between Earth and Mars fluctuating between 6 and 45 minutes depending on the planetary constellation. Intermittent connectivity, limited bandwidth, and asymmetrical communication system capabilities between ground stations and spacecraft further challenge the communication. Standard Internet protocols, specifically the Transmission Control Protocol (TCP), have been found unsuitable in this context [16]. Delay-tolerant Networking (DTN) protocols, such as the Bundle Protocol [7], have been developed since early 2000 to address the constraints of space communication. Since then, developments to replace TCP with QUIC have sparked a new effort in the IETF working group TIPTOP to reuse Internet protocols and software in space [6]—an interplanetary Internet [20] may be less fiction than potential reality in the future.

Evaluating protocols and applications under realistic conditions has proven beneficial since the invention of the Internet. This, however, remains difficult in space scenarios due to the high technical and logistical barriers when deploying necessary infrastructure. So far, no representative communication data set for this environment was published, which hinders empirical evaluation. Common guidelines on deep space link characteristics to use in evaluation do not appear to be available either, as manifold interactions of the authors with engineers of the space networking community suggest.

In this paper, we provide the first analysis of the NASA Deep Space Network (DSN), one of the primary systems used for deep space communication [28]. DSN supports networking for some of the most important space missions from the Voyager spacecraft to the Artemis II moon mission. Based on the telemetry data from these space missions, we examine latency, data rates, and connectivity disruptions for different groups of spacecraft over 11 months. We continue collecting the DSN data set and make historical data publicly available.

In § 2, we present background and related work. § 3 describes our method for collecting deep space communication data. We discuss the results of our analysis in § 4. Finally, we conclude and outline possible areas for future work in § 5.

2 Background and Related Work

This section presents the unique challenges of deep space communication and illustrates relevant deployment scenarios. We introduce the NASA Deep Space Network (DSN), the foundation of our analysis, and discuss related work.

2.1 Networking in Space

Scenarios. Current space networks rely primarily on Earth-centric point-to-point links, with Moon and Mars surface missions occasionally utilizing orbiters as telemetry relays. Future near-term lunar architectures will likely be based on interoperable, multi-node DTN topologies. This shared infrastructure will integrate landers, orbiters, and relays at Earth-Moon Lagrange Point 2 (L2). Mars networks are expected to follow similar architectures [13, 34].

Beyond planetary deployments, dense spacecraft clusters operate at the gravitationally balanced Earth-Sun Lagrange

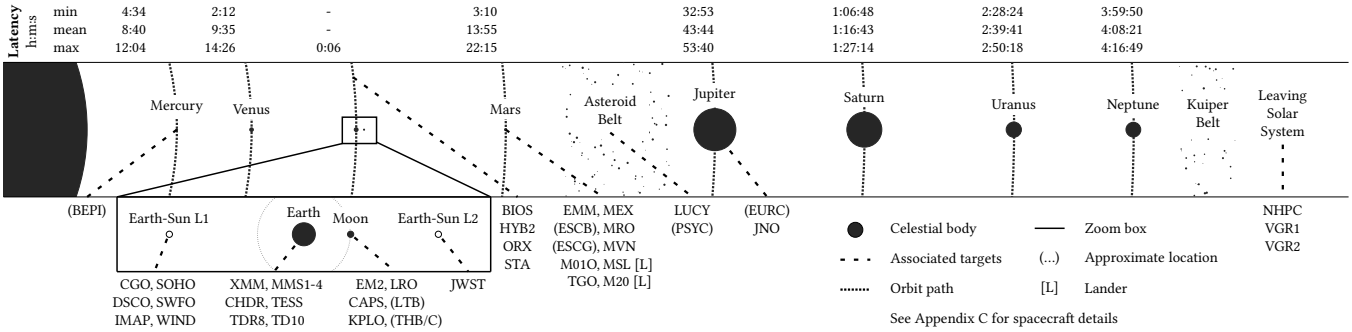


Figure 1: DSN targets (*aka* spacecraft) and their locations in space. Distances are not to scale, but planet sizes are. The one-way light times from Earth to other planets (and to the Earth-Sun Lagrange points) are shown (top).

points (L1/L2). These stable orbital positions guarantee continuous line-of-sight and minimal latency variance [15].

Unlike these stable deployments, missions in varying heliocentric orbits act as highly dynamic nodes with continuously shifting propagation delays. At the extreme network edge, interstellar probes include Voyager and New Horizons.

Challenges. Space networks face several challenges. First, propagation delays are significantly larger compared to terrestrial networks. One-way latencies for distant probes such as Voyager 1 [29] can approach a full day.

Second, space communication relies on radio or, more recently, optical links [21], both requiring line of sight, making them vulnerable to frequent disruptions from planetary rotation or obstruction by celestial bodies. Environmental factors, *e.g.,* signal-altering solar winds or atmospheric attenuation, can significantly degrade the physical layer further, decreasing achievable bandwidths [32]. Beyond physical line-of-sight interruptions, links face severe resource contention. Earth-bound infrastructure lacks the capacity to concurrently serve the connectivity requirements of all active missions, leading to scheduled disruptions [16].

Third, thermal vacuum severely restricts power dissipation, imposing strict hardware limitations to space missions. Additionally, cosmic radiation mandates radiation-hardened components [3], resulting in processing hardware that operates up to two orders of magnitude slower than terrestrial counterparts [25]. Consequently, available bandwidth is severely limited, *e.g.,* Voyager 1 and 2 rely on a 160 bit/s downlink and a 16 bit/s uplink [26]. This asymmetry complies with operational requirements, as uplinks only need to deliver compact command sequences whereas downlinks must return substantial telemetry and scientific payloads [17].

2.2 NASA Deep Space Network

The Deep Space Network (DSN), operated by NASA, consists of Deep Space Communication Complexes (DSCC) located in Goldstone (USA), Madrid (Spain), and Canberra (Australia). These complexes are spaced evenly apart in longitude around Earth, enabling smooth transition from site to site when communicating with a spacecraft as the Earth rotates [28].

Each DSCC consists of multiple antennas, with dishes measuring either 34 m or 70 m in diameter. Each DSCC has one 70 m antenna to achieve communication at larger distances [23]. In addition, Goldstone and Canberra each operate three 34 m antennas, while Madrid operates five. Thus, there are in total 14 antennas in service for the DSN.

The DSN antennas can operate in multiple frequency bands in parallel. Different dishes support different bands, with upgrades adding support for higher frequencies [23].

Figure 1 provides an overview of the spacecraft in communication with the DSN and their location in the solar system during our observation period. At the top of the figure, we supplement latency estimates for the next ten years calculated using NASA NAIF's SPICE Toolkit [1, 2, 4].

2.3 Related Work

With the introduction of Starlink and other planned mega-constellations around Earth, most recent research focuses on Low Earth Orbit (LEO) satellites [10, 22, 27]. LEO constellations face similar but distinct challenges regarding latencies and disruptions compared to deep space communication. Latencies are high compared to the terrestrial Internet but low compared to a deep space link. Disruptions may appear as well, *e.g.,* when no satellite is above the horizon for a particular ground station [24, 35]. Inter-satellite communication on a scale of mega-constellations will most likely not matter to the deep space scenario for the foreseeable future [13, 34].[1]

The analysis of deep space networking is often based on simulations, relying on synthetic data [33]. Our work can serve as realistic input to such simulations.

[1]Networking via geostationary satellites yields higher latencies but those also amount only to a fraction of deep space links; moreover, the fixed satellite positions relative to the Earth surface exhibit rather stable links, so they bear little similarity to a deep space network.

Prior research on DSN analyzes the effects of aging ground infrastructure and increasing demand, highlighting resource shortages [31], or focuses on the operational issues such as scheduling [11, 19] and improvements to the physical layer, especially considering advancements in optical communication [12, 14]. We contribute insights into the operating environment, relevant to design space communication.

The challenges in creating a solar system Internet were first investigated in the early 2000 [16]. Since then, Delay-tolerant Networking has been introduced [8, 9] to address these problems. We want to expand upon and quantify some of the initial and often cited claims regarding (*i*) low data rates, (*ii*) high latencies, and (*iii*) intermittent connectivity.

3 Method

Gathering deep space communication data is a challenge on its own, since operating the infrastructure is costly. We rely on data sources offered by the spacecraft and network operators. The NASA Jet Propulsion Laboratory provides metadata for DSN communication through the DSN Now website [30]. The information is sourced directly from the ground stations of the DSN and updated every 5 seconds. To allow for longitudinal analysis, we scrape this website since 2025-05-27 and make an archive of the data set available.

The DSN Now data is structured hierarchically. At each point in time a DSN antenna can have multiple target spacecraft that represent either a satellite, a lander, or a rover. Each target communicate via multiple uni-directional signals. Which data points are present per signal depends on the direction of communication with differences between uplinks (from antenna to target) and downlinks (from target to antenna). Data about a target and signal is not reported when there is no contact between the target and the DSN.

We complement the DSN Now data by manually curating location descriptors to group spacecraft based on the location they target (see § 2.1). The groups consist of spacecraft orbiting Mercury, Earth, the Moon, Mars, Jupiter, Sun-Earth Lagrange points, or spacecraft on a trajectory to escape the solar system. This does not cover all spacecraft. Special cases include spacecraft orbiting the Sun that follow complex trajectories not easily described in one word without extensive background knowledge. With this classification, we can differentiate spacecraft even when they show otherwise similar behavior.

3.1 Data Quality

The nature of this data demands special attention related to completeness, precision, and interpretation. When processing the data set first, we found some issues.

Specific data points show erratic behavior in certain scenarios. For example, the reported received power suddenly drops by around 300 dBm, a value, which is physically implausible.

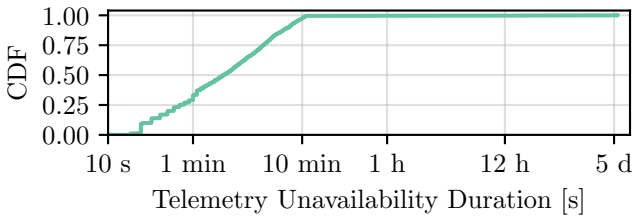


Figure 2: CDF of DSN telemetry unavailability showing many individual events below ten minutes.

Data points including the round-trip time were sometimes missing while others were reported. Others, such as the signal frequency, never contained data at all. Unfortunately, this also applies to the data rate of uplink signals. Even when data is present, the precision is often low. Data points are rounded, with some only including three decimal digits. The temporal resolution of five seconds is too low to capture communication characteristics such as sudden traffic bursts.

Still, even when considering these issues, DSN Now remains the best source for deep space communication available to us, as it is the only public source containing live information of a relevant number of diverse spacecraft.

3.2 Data Sanitization

To increase the usability of the DSN Now data, thorough data sanitization measures are required. We focus on target round-trip times, signal data rates, and information gained from the observation over time, such as the disruption of connectivity. The following describes the measures we took to sanitize these data points.

DSN Now describes the current usage of the dish via the dish activity data point, containing values indicating radio astronomy, maintenance, testing, or most importantly, *Spacecraft Telemetry, Tracking & Command*. We observed spacecraft communication only during the last activity. Others, such as tests, often cause inexplicable data anomalies. A similar data point exists to indicate the state of a signal. If this signal activity is indicated as *false*, the signal might be briefly disrupted or in the setup or teardown process. Hence, we filter the data of the related dish and signal activity.

To exclude spacecraft that did not exist at the time they showed up in DSN Now, we cross-verify these dates with actual launch dates . This proved to be effective, as the DSN Now data mentions a Blue Origin spacecraft only to be launched one year later. We assume that these artifacts stem from preliminary testing that propagates erroneously to DSN Now.

Our location descriptors are static and should not apply to spacecraft while they are in transit. We avoid misclassification using a second date that indicates arrival of a satellite

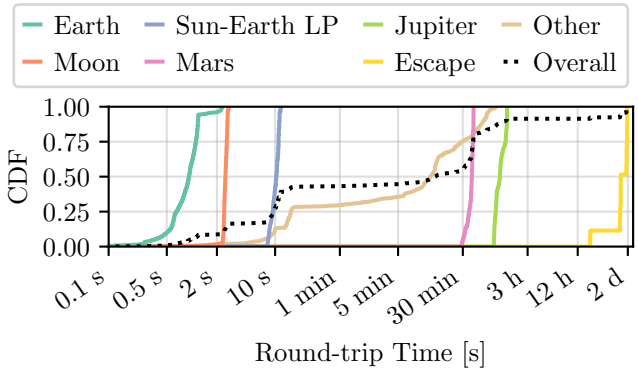


Figure 3: Round-trip time distributions for different target locations compared to the overall data set.

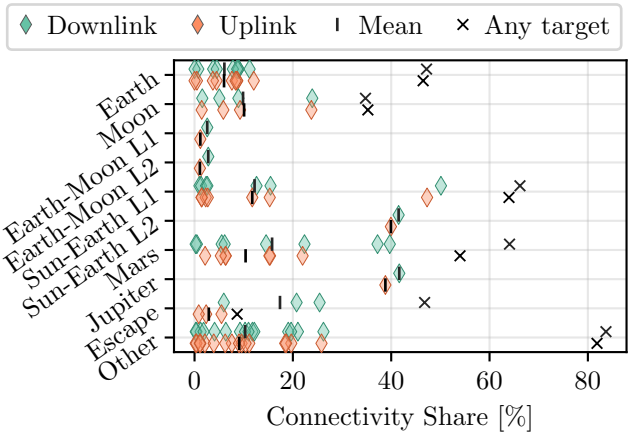


Figure 4: Per target up- and downlink connectivity share, mean and aggregate grouped by target location.

at its destination. Prior to reaching their final destination, spacecraft are assigned the *Other* category.

To analyze connectivity disruptions, we need to differentiate them from unavailability of the DSN Now website and artifacts of our scraping. We define disruptions as the time between data points for any active signals of the same target. We define telemetry unavailability as periods for which we do not possess updated DSN Now data caused by, *e.g.*, website maintenance or scraping issues. The lower bound of the telemetry unavailability time should be close to the DSN update interval of five seconds, the DSN Now issued timestamp shows variations of up to seven seconds between individual updates. However, due to our scraping script not compensating for marginally higher than expected processing time, we occasionally missed single updates. As a result, our update interval, and thus our lower bound for telemetry unavailability, is 15 seconds.

There are numerous telemetry unavailability events, totaling a duration of 19 days (see Figure 2). Thirteen days can be attributed to four day-long periods due to DSN Now maintenance, and two days due to downtime of our measurement VM. We aim to provide a lower bound on the amount of disruptions. Hence, we discard any disruption that precisely overlaps telemetry unavailability and disruptions spanning around telemetry unavailability lasting longer than one day.

4 Results

We investigate DSN Now data regarding latency (§ 4.1), connectivity (§ 4.2), disruptions (§ 4.3), and data rates (§ 4.4). We will present results from 2025-05-27 to 2026-04-28.

4.1 Latency

Average latencies can be easily predicted due to direct correlation between distance and latency in line-of-sight communication. Therefore, we focus on the distribution of round-trip times for different locations of targets (see Figure 3). Concentrations in the overall round-trip time distribution correspond to the fraction of communication with the respective target locations.

Approximately one-quarter of the overall distribution reflects DSN communication with Mars spacecraft. Mars is a unique location in that there are many spacecraft closely grouped at a relatively high distance. The low angular separation in the sky above an antenna enables parallel downlinks from multiple targets. This may have contributed to the high share compared to other locations. The round-trip time during our observation period ranges from 29 to 40 minutes, which is high in the context of the expected values during the next ten years (see Figure 1).

The next most relevant distinct group are spacecraft located around the Sun-Earth Lagrange points. With a mean round-trip time of 10 s and deviations of 2 s in either direction, they amount to 20% of the overall distribution. Focusing on Mars *and* Moon [5] in space networking research may thus be an oversimplification, as the latencies differ by orders of magnitude compared to the Sun-Earth Lagrange points.

Spacecraft whose location we could not classify account for a fourth of communication, while the rest is split almost evenly between Earth, Moon, Jupiter, and spacecraft on a trajectory to escape the solar system.

4.2 Connectivity

Figure 4 demonstrates that most spacecraft were not in contact with the DSN for most of the time. The connectivity share, the fraction of time a target had an active signal to or from the DSN, varies with the target location. Mars spacecraft had a mean connectivity share of 16 % in the downlink while their uplinks were only available 10 % of the time.

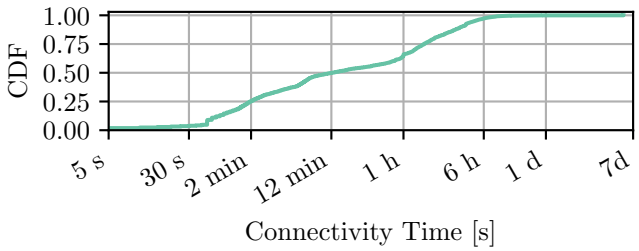


Figure 5: Distribution of the connectivity time.

Moon spacecraft up- and downlinks both exhibit a connectivity share of 10 % on average, not taking into account the close-by Earth-Moon Lagrange points.

Considering the theoretical case that there was a planetary network on these bodies, the connectivity share to that network would have been 35 % for the Moon. The share to and from Mars would have been considerably higher at 55 % and 65 %, respectively. The Sun-Earth Lagrange point L1 would have had downlink connectivity above 66 % of the time, with only a minor difference to its uplinks.

The duration of each contact between the DSN and a target varies significantly, see Figure 5. We define the connectivity time as the duration a spacecraft in the DSN retains an active signal and analyze this connectivity time for all spacecraft. Most contacts last between 50 seconds and 8 hours (1st–99th percentile). SOHO achieved the longest undisrupted period of 5 days and 15 hours. However, half of all contacts terminate within 12 minutes, revealing highly intermittent link conditions.

In an ideal scenario, the globally distributed DSN stations could provide continuous connectivity through handovers as the Earth rotates. This capability was demonstrated during the crewed Artemis II Moon mission (Figure 6). However, such extensive coordination remains exceptional and is rarely used for typical missions. Short, minute-long disruptions still occurred when both concurrently used dishes reported inactive signals.

4.3 Disruptions

The connectivity of a spacecraft to the DSN can be disrupted for multiple reasons. DSN Now does not provide information on why a disruption occurred. Therefore, we examine the dish behaviors in between the last active and new signal of a spacecraft. In case the new signal is linked to a different dish, we consider only the hour following the beginning of the disruption. We apply the following heuristics to derive a reason for the disruption:

Inactive signal The dish keeps a signal to the spacecraft but the signal is reported as inactive.

Other dish activity The dish is used for other tasks, such as astronomy, or is undergoing maintenance.

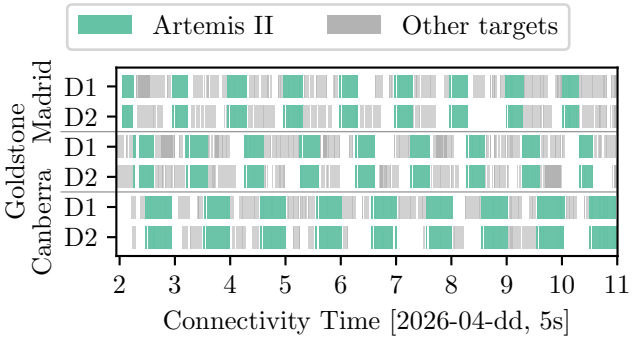


Figure 6: Artemis II connection timeline with 2 dishes per DSN complex, gray showing usage by other targets.

Other target At least one other target is tracked.

Earth rotation The disruption begins when the dish elevation angle was low ($< 15^\circ$) and resumes at a new dish or after at the earliest 12 hours of downtime.

The distribution of the disruption times and the derived reasons are shown in Figure 7. The station logically coming into line of sight as the Earth rotates is the next station. We can observe that disruptions ending at the next station typically last less than eight hours. Also at the next station, a few minute-long disruptions occur, indicating an effort to perform a smooth handover between stations. Around every eight hours of additional disruption time, the station next in the cycle sees a peak in resumed connectivity. Disruptions longer than a day are rare.

In the case that the same station is used before and after the disruption, we find the typical time to recover to be around 30 minutes long and very few disruptions lasting longer than an hour. In that range, signal inactivity is the predominant reason for the disruption. Only Mars spacecraft cause the "other target" heuristic to apply for these shorter disruptions. However, even before Earth rotation, it is the most common reason for disruptions longer than an hour.

As our data set does not contain scheduling information, we cannot decisively tell whether an intended prioritization or an operational issue was the cause for an "other target" disruption. We suggest the following reasons in the case of two spacecraft in close proximity. First, the dish might be performing time-division multiplexing, as it is able to communicate with either target without changing its pointing direction. Second, the dish tracking might have falsely locked onto an unintended spacecraft.

The heuristic "other dish activity" appears more often and without a clear pattern. In fewer than 500 occurrences, our heuristics either classified multiple reasons or none at all. One cause is the deliberate one-hour cut-off in case the dish changed after a disruption, leading to either not enough or too much data being considered.

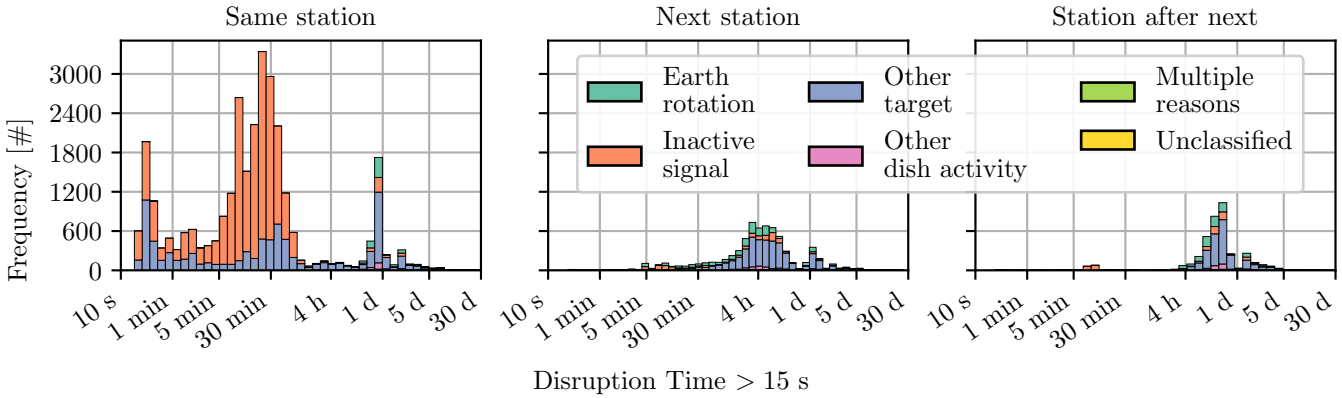


Figure 7: Distribution of disruption time and presumed reason for the disruption. Resumption with another DSN station is shown separately. Our scraping interval limits the minimum disruption time.

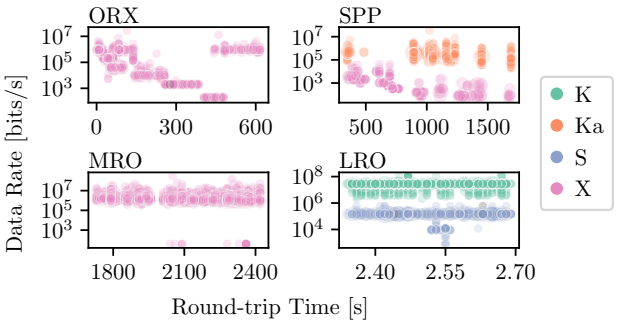


Figure 8: Relationship between data rate, round-trip time, and frequency band for two fast traveling (ORX, SPP), one Mars (MRO), and one Moon (LRO) spacecraft.

The many disruptions belonging to the heuristic "other target" allow the following conclusions. First, to ensure disruption-free operation, the DSN would need significantly increased capacity. Second, more capable networking infrastructure in popular locations such as Mars could alleviate load on Earth ground stations and lead to less disruptions.

4.4 Data Rates

Data rates vary greatly between targets, see Figure 8. Only spacecraft closest to Earth can achieve very high bandwidths. However, for higher latencies, data rates do not always degrade as maybe expected. To mitigate the impact of heterogeneous onboard technologies among different target groups, we investigate spacecraft that traveled relatively large distances during our observation period. OSIRIS-REx (ORX) partly demonstrates the expected behavior up until there is a sudden increase in data rate not explainable using data points available to us. For the Parker Solar Probe (SPP), a direct correlation exists for the X band but not for the higher-frequency Ka band. We suggest that the effects of latency on data rate are in many circumstances compensated for at the lower layers. We present the same analysis for all targets of the DSN in the appendix, Figure 16.

5 Conclusion and Outlook

In this paper, we collected and analyzed telemetry data from DSN Now to characterize, for the first time, the communication patterns of the Deep Space Network over eleven months. Our latency analysis reveals predictable distance-latency correlations forming distinct delay clusters, with nearly half of the traffic split between distant Mars and closer Lagrange-point missions. Regarding connectivity, we show that while the DSN is technically capable of undisrupted multi-day communication, practical connections are typically short, with half lasting under 12 minutes. Furthermore, frequent disruptions from competing target scheduling emphasize the need for DSN capacity increases. Finally, while peak bandwidths are exclusive to near-Earth missions, overall data rates do not strictly degrade with latency due to compensating lower-layer mechanisms and frequency choices. Future DSN deployments could improve Lunar and Mars connectivity while reducing DSN scheduling loads, ultimately resulting in fewer disruptions.

Our data collection, which we will make publicly available, includes time series of ground station, dish, and target identifiers together with RTT, up/downlink signal, and the observed downlink bitrate. When complementing the downlink data rates with plausible assumptions for the uplink, *e.g.*, based upon frequency band, transmission technology, and distance, the data set allows parameterizing DSN simulations or emulation tools for future evaluation. Our findings may serve as complementary guidance for deep space protocol design and evaluation in the future.

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A Ethics

This work does not raise any ethical issues.

B Additional Details

B.1 Mars and Moon Details

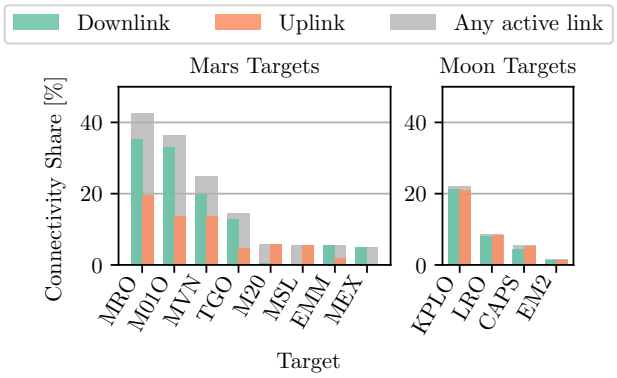


Figure 9: Connectivity share of Mars and Moon spacecraft over the entire observation period.

Figure 9 provides per-target analysis of the connectivity share around Mars and Moon. Mars downlinks are more often available than the respective uplinks. Additionally, Mars targets achieve a higher connectivity share compared to Moon targets.

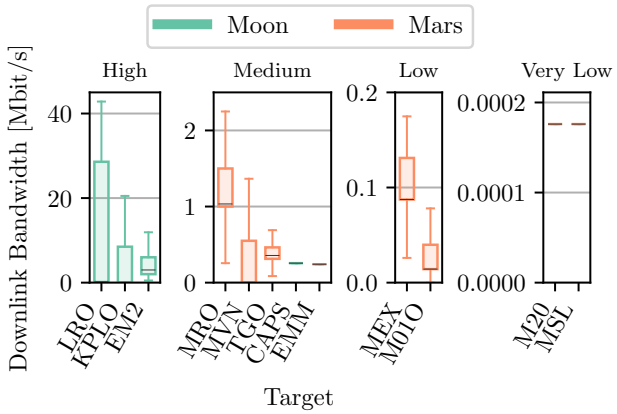


Figure 10: Distribution of downlink bandwidth across different Moon and Mars targets.

Moon targets utilize higher downlink data rates compared to Mars targets as can be seen in Figure 10. There are several orders of magnitude of difference between the targets. Individual targets may experience significant fluctuations in data rate.

B.2 Stability

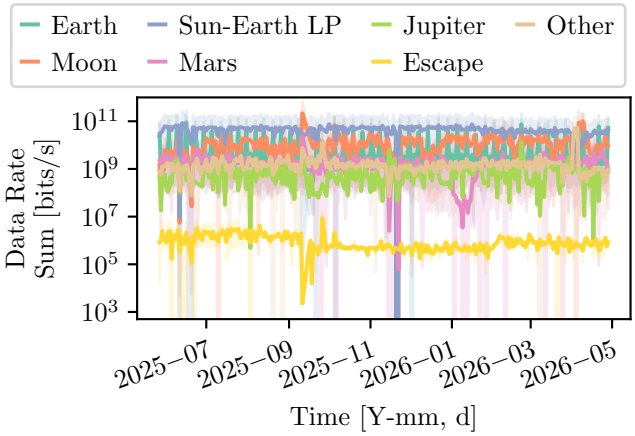


Figure 11: Sum of data rate per target location, binned per day.

Data rates (see Figure 11) and latency (see Figure 12) varied little over our observation period. A drop in data rates in September for spacecraft on an escape trajectory can be explained by the large dish "DSS-14" being taken offline after being damaged [18]. Large dishes are crucial for communication with the Voyager spacecraft.

Figure 13 demonstrates the stability of the disruption distribution over ten slices of our observation window. Each line represents a slice containing 10% of our observed data. Slices are non-overlapping and sequential.

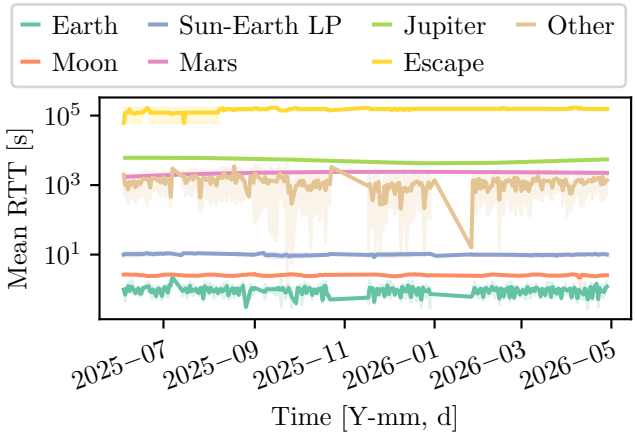


Figure 12: Mean of round-trip time per target location, binned per day.

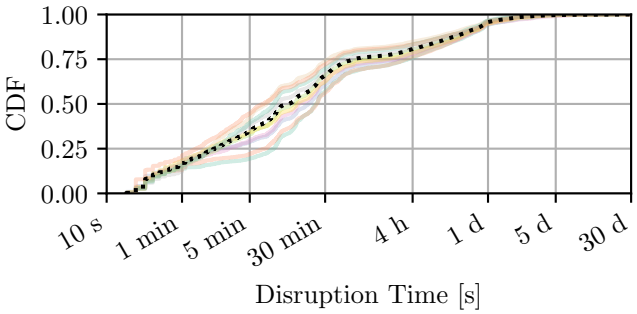


Figure 13: Stability of the disruption distribution. Lines represent 10% of the observation window.

B.3 Correlating Data Rates

Figure 14 correlates downlink data rates with the launch date of the target spacecraft. Contrary to our expectation, there is no general increase in data rate over time beyond the start of the millennium. Voyager 1 and 2, which were launched in 1977 and are further away than any other target, explain the lower data rates for the 1970s group.

Figure 15 shows per-location distributions of downlink data rates, as well as an overall distribution. In the case of multiple concurrent signals from a target, we sum their data rates. Spacecraft around the closest locations, Earth, Moon, and the Sun-Earth Lagrange Points, generally show higher data rates than those at further away locations.

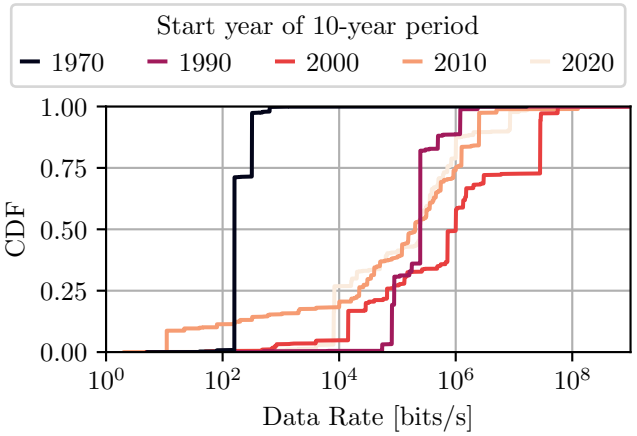


Figure 14: Data rate distribution for different target launch year groups.

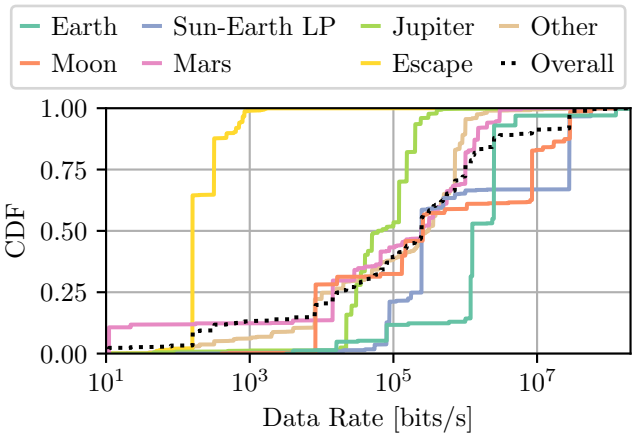


Figure 15: Data rate sum distributions for different target locations compared to the overall data set.

B.3.1 Per-Target Round-trip Time and Data Rate. Figure 16 depicts the relationship of round-trip time and downlink data rates for all DSN targets. With increasing round-trip time, data rates either decrease (LUCY, ORX) or do not correlate (ACE, STA). Satellites supporting different frequency bands can show independent behavior per band (SPP).

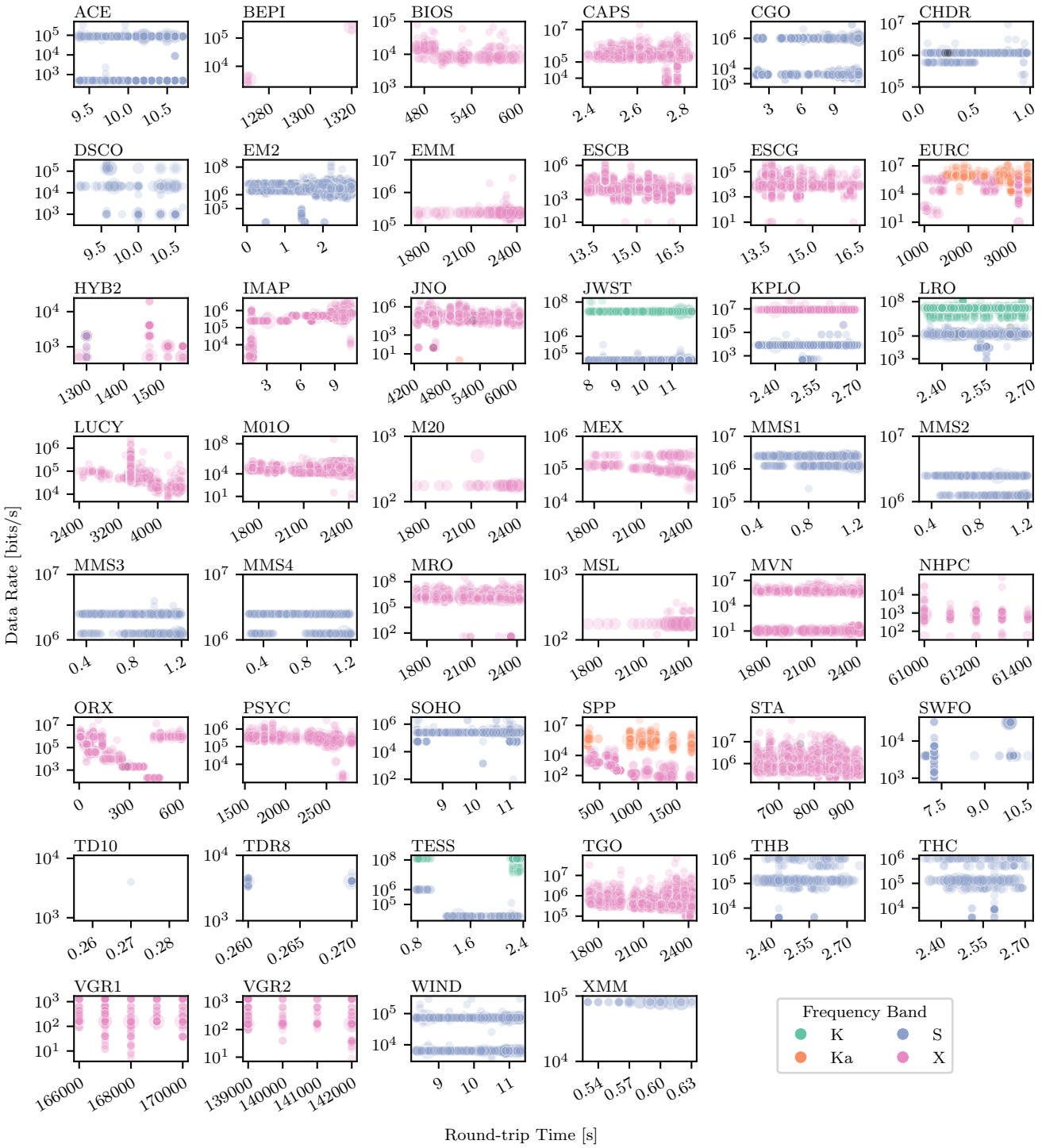


Figure 16: Relationship between data rates and round-trip time split by target.

B.4 Per-Location Disruption Reasons

We split our analysis from § 4.3 per target instead of station in Figure 17. Notably, only Mars spacecraft experience sub-hour disruptions that can be attributed to our "other target" heuristic. Lagrange points L1 and L2 exhibit comparable behavior in the Sun-Earth, as well as the Earth-Moon system.

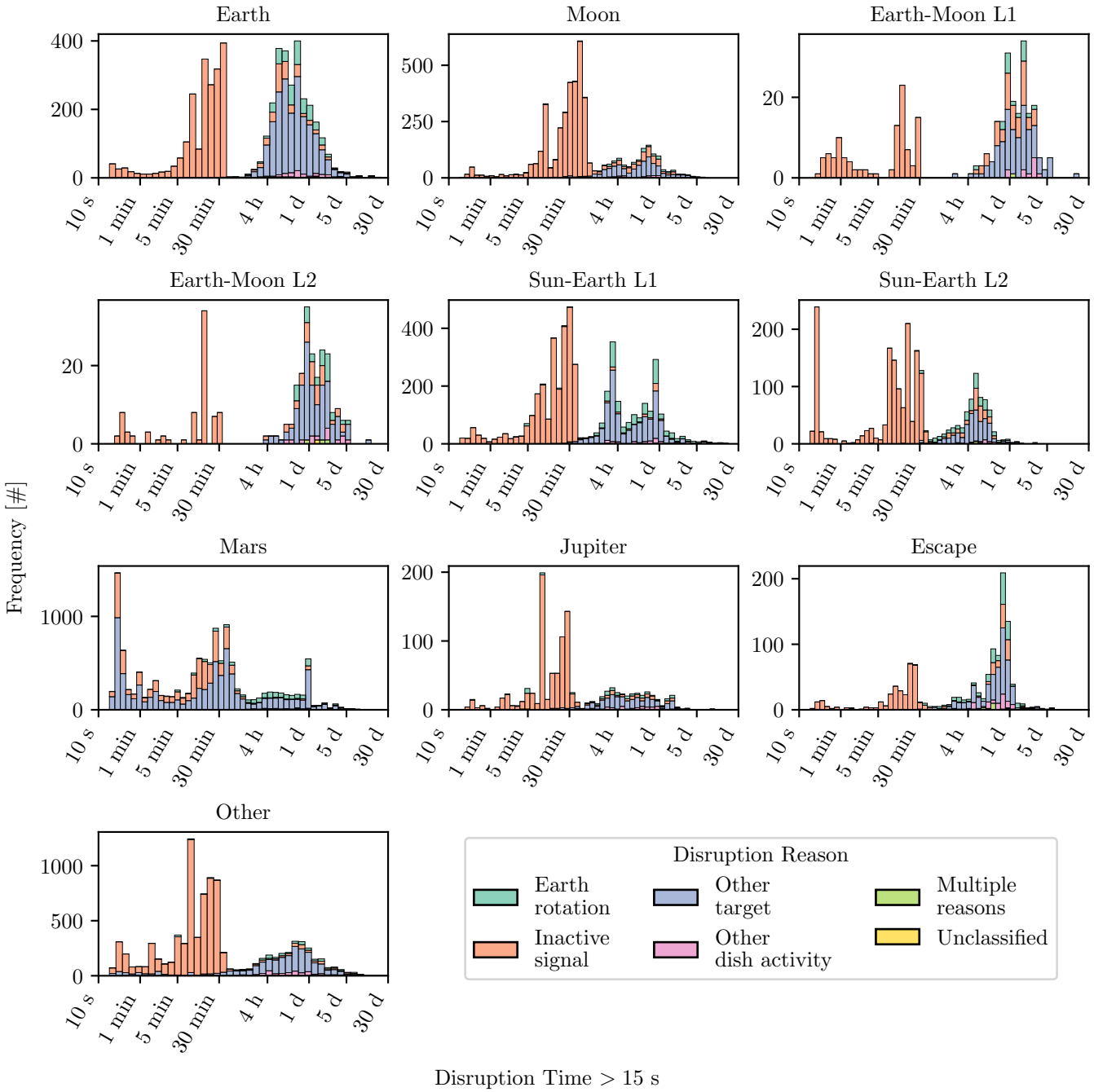


Figure 17: Disruption time distribution and reasons split by the target location.

C DSN Now Targets

Table 1: Information about DSN targets. Dates are sourced mostly from NASA's informational websites (exact source can be found in our artifacts). Dates set in the future are estimates and subject to change. Especially the estimated arrival date is to be used with caution.

ID	Name	Launch	Est. arrival	Type	Destination
ACE	Advanced Composition Explorer	1997-08-25	1998-07-21	Satellite	Sun-Earth L1
AGM1	Astrobotic Griffin Lander	2026-07-31	-	Lander	Moon
B101	Blue Origin Blue Moon Human Landing System MK1 SN101	-	-	Lander	Moon
BEPI	BepiColombo	2018-08-20	2026-11-01	Satellite	Mercury
BIOS	BioSentinel	2022-11-16	2022-12-05	Satellite	Other
CAPS	CAPSTONE	2022-06-28	2022-11-13	Satellite	Moon
CGO	Carruthers Geocorona Observatory	2025-09-24	2026-01-08	Satellite	Sun-Earth L1
CHDR	Chandra	1999-07-23	1999-08-07	Satellite	Earth
DSCO	Deep Space Climate Observatory (DSCOVR)	2015-02-11	2015-06-08	Satellite	Sun-Earth L1
EM2	Human Space Flight: Artemis II	2026-04-01	2026-04-06	Satellite	Moon
EMM	Emirates Mars Mission (Hope)	2020-07-20	2021-02-09	Satellite	Mars
ESCB	ESCAPADE Blue	2025-11-13	2028-06-01	Satellite	Mars
ESCG	ESCAPADE Gold	2025-11-13	2028-06-01	Satellite	Mars
EURC	Europa Clipper	2024-10-14	2030-01-01	Satellite	Jupiter
HYB2	Hayabusa-2	2014-12-03	2018-06-27	Satellite	Other
IMAP	Interstellar Mapping and Acceleration Probe	2025-07-24	2026-02-01	Satellite	Sun-Earth L1
JNO	Juno	2011-08-05	2016-07-04	Satellite	Jupiter
JWST	James Webb Space Telescope	2001-12-25	2022-01-24	Satellite	Sun-Earth L2
KPLO	Korea Pathfinder Lunar Orbiter	2022-08-05	2022-12-26	Satellite	Moon
LRO	Lunar Reconnaissance Orbiter	2009-06-18	2009-06-23	Satellite	Moon
LTB	Lunar Trailblazer	2025-02-26	-	Satellite	Moon
LUCY	Lucy	2021-10-16	2023-11-01	Satellite	Other
M01O	Mars Odyssey	2001-04-07	2001-10-24	Satellite	Mars
M20	Mars 2020 (Perseverance)	2020-07-30	2021-02-18	Rover	Mars
MEX	Mars Express	2003-06-02	2003-12-01	Satellite	Mars
MMS1	Magnetospheric Multiscale Mission 1	2015-03-12	2015-03-18	Satellite	Earth
MMS2	Magnetospheric Multiscale Mission 2	2015-03-12	2015-03-18	Satellite	Earth
MMS3	Magnetospheric Multiscale Mission 3	2015-03-12	2015-03-18	Satellite	Earth
MMS4	Magnetospheric Multiscale Mission 4	2015-03-12	2015-03-18	Satellite	Earth
MRO	Mars Reconnaissance Orbiter	2005-08-12	2006-03-10	Satellite	Mars
MSL	Mars Science Laboratory (Curiosity)	2011-11-16	2012-08-06	Rover	Mars
MVN	MAVEN	2013-11-13	2014-09-21	Satellite	Mars
NHPC	New Horizons	2006-01-19	2015-07-14	Satellite	Escape
ORX	OSIRIS-REx	2016-09-08	2020-08-20	Satellite	Other
PSYC	Psyche	2023-10-13	2029-07-01	Satellite	Other
RST	Roman Space Telescope	2027-05-01	-	Satellite	Sun-Earth L2
SOHO	SOHO	1995-12-02	1996-12-02	Satellite	Sun-Earth L1
SPP	Parker Solar Probe	2018-08-12	2021-12-14	Satellite	Other
STA	STEREO A	2006-10-26	2006-10-26	Satellite	Other
SWFO	Space Weather Follow On-Lagrange 1	2025-09-24	2026-01-23	Satellite	Sun-Earth L1
TD10	TDRS 10	2002-12-05	2002-12-05	Satellite	Earth
TDR8	TDRS 8	2000-06-30	2000-06-30	Satellite	Earth
TESS	Transiting Exoplanet Survey Satellite	2018-04-18	2018-07-25	Satellite	Earth
TGO	Trace Gas Orbiter	2016-03-14	2016-10-19	Satellite	Mars

(Cont.)

ID	Name	Launch	Est. arrival	Type	Destination
THB	THEMIS B	2007-02-17	2010-08-25	Satellite	Earth-Moon L2
THC	THEMIS C	2007-02-17	2010-10-22	Satellite	Earth-Moon L1
VGR1	Voyager 1	1977-09-05	1980-11-09	Satellite	Escape
VGR2	Voyager 2	1977-08-20	1981-08-25	Satellite	Escape
WIND	WIND	1994-11-01	1996-11-01	Satellite	Sun-Earth L1
XMM	X-ray Multi-Mirror Mission (XMM-Newton)	1999-12-10	1999-12-10	Satellite	Earth

Table 2: Target count per destination (excludes targets not launched as of writing). Actual counts in our analysis may be lower as targets are assigned to "Other" before their arrival date. Locations may be merged when separate results would be visually indistinguishable or not meaningful due to representing only a single target.

a Raw destinations

Destination	Targets [#]
Earth	9
Moon	6
Earth-Moon L1	1
Earth-Moon L2	1
Sun-Earth L1	7
Sun-Earth L2	1
Mercury	1
Mars	10
Jupiter	2
Escape	3
Other	7

b Merged destinations

Destination	Targets [#]
Earth	9
Moon	8
Sun-Earth LP	8
Mercury	1
Mars	10
Jupiter	2
Escape	3
Other	7